

GREENOCK MRCC, United Kingdom

WMO: 031380

Lat: 55.967N

Lon: 4.800W

Elev: 5

StdP: 101.26

Time Zone: 0.00 (EUW)

Period: 90-07

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.2	0.5	-5.4	2.4	2.4	-3.9	2.7	3.2	20.4	5.6	18.4	6.6	3.2	120	0.687

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.0	22.1	16.6	20.6	16.0	19.1	15.0	17.7	21.0	16.7	19.2	15.9	18.3	3.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.4	11.7	18.9	15.6	11.1	18.2	14.8	10.5	17.4	49.9	21.3	46.9	19.1	44.6	18.4	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.3	14.1	12.3	-1.8	25.3	0.9	1.5	-2.5	26.4	-3.0	27.2	-3.5	28.1	-4.2	29.2		
(5)			-3.0	19.5	1.4	2.0	-4.0	20.9	-4.8	22.1	-5.6	23.2	-6.6	24.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	5.7	5.6	6.8	8.9	11.6	13.7	15.5	15.5	13.3	10.5	8.0	5.7
	DBStd	4.31	2.48	2.26	2.22	2.23	2.28	2.07	2.07	2.03	1.96	2.47	2.49	2.88
	HDD10.0	642	134	124	101	47	9	1	0	0	1	26	67	133
	HDD18.3	3020	393	358	357	284	208	141	92	93	151	244	309	390
	CDD10.0	673	1	1	2	13	59	111	169	169	101	40	8	1
	CDD18.3	8	0	0	0	0	0	1	3	4	1	0	0	0
	CDH23.3	23	0	0	0	0	1	4	8	9	1	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	5.4	6.7	6.2	6.3	5.4	4.8	4.7	4.7	4.5	4.7	5.2	5.6	5.7
Precipitation	PrecAvg	1532	177	132	132	91	83	85	100	117	122	164	159	171
	PrecMax	1895	267	255	233	150	166	150	165	254	212	274	304	295
	PrecMin	1165	47	35	61	49	18	36	30	32	35	54	60	45
	PrecStd	199	58	63	43	25	39	30	32	55	47	65	59	75
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.0	11.8	12.9	18.0	21.5	23.4	24.6	24.6	20.7	16.6	13.7	12.5
		MCWB	10.5	10.0	9.7	12.8	14.6	15.5	18.5	18.1	15.9	13.7	11.9	11.1
	2%	DB	10.9	10.6	11.7	15.1	18.5	20.1	22.1	22.0	18.6	15.4	12.9	11.5
		MCWB	9.7	8.9	9.4	10.8	13.5	14.8	17.1	16.5	14.9	13.3	11.3	10.1
	5%	DB	10.2	9.9	10.8	13.2	16.8	18.3	20.6	20.1	17.3	14.5	12.3	10.7
		MCWB	8.9	8.3	8.8	9.9	12.4	13.7	16.3	16.2	14.2	12.6	10.9	9.3
	10%	DB	9.3	9.0	10.0	12.1	15.3	17.0	19.0	18.5	16.2	13.7	11.6	9.8
		MCWB	8.0	7.4	8.2	9.4	11.5	13.2	15.3	15.2	13.7	11.9	10.2	8.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.7	10.2	11.2	13.7	15.9	17.3	19.5	19.0	16.7	14.5	12.5	11.5
		MCDB	11.5	11.1	11.8	17.1	19.5	20.7	22.9	22.7	19.1	16.0	13.2	12.3
	2%	WB	10.0	9.3	10.0	11.8	14.2	15.8	17.8	17.8	15.8	13.8	11.8	10.4
		MCDB	10.8	10.3	11.1	13.8	17.5	18.9	21.2	21.1	17.6	15.1	12.6	11.4
	5%	WB	9.1	8.5	9.3	10.7	13.1	14.7	16.8	16.8	15.0	13.0	11.1	9.4
		MCDB	10.0	9.6	10.5	12.6	15.9	17.2	19.5	18.9	16.6	14.2	12.1	10.5
	10%	WB	8.1	7.6	8.5	9.8	12.2	13.8	15.9	15.9	14.1	12.1	10.3	8.5
		MCDB	9.2	8.8	9.8	11.7	14.6	16.1	18.4	18.1	15.8	13.5	11.5	9.6
Mean Daily Temperature Range	5% DB	MDBR	3.3	3.6	4.1	4.9	5.7	5.2	5.0	5.0	4.5	3.6	3.4	3.1
		MCDBR	4.1	4.2	4.9	7.2	8.8	8.0	7.9	8.0	6.0	4.3	3.9	4.2
		MCWBR	4.1	3.7	3.4	4.3	4.8	4.1	4.2	4.0	3.6	3.2	3.8	4.2
	5% WB	MCDBR	4.0	4.0	3.9	6.0	7.5	6.7	6.8	6.3	5.5	3.7	3.8	4.1
		MCWBR	4.2	3.8	3.5	4.3	4.7	4.3	4.1	3.9	4.3	3.5	4.1	4.2
Clear-Sky Solar Irradiance	taub	0.337	0.347	0.371	0.387	0.391	0.385	0.387	0.371	0.363	0.346	0.327	0.321	
	taud	2.196	2.296	2.277	2.276	2.314	2.364	2.366	2.422	2.441	2.413	2.288	2.173	
	Ebn at Noon	539	708	784	830	849	858	848	841	793	707	558	455	
	Edh at Noon	59	78	100	115	118	114	111	99	85	69	53	48	
All-Sky Solar Radiation	RadAvg	0.46	1.11	2.20	3.61	4.66	4.56	4.36	3.50	2.42	1.30	0.60	0.34	
	RadStd	0.07	0.14	0.18	0.28	0.43	0.50	0.44	0.28	0.19	0.18	0.06	0.04	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page